TIME-OF-FLIGHT DISTANCE SENSOR

ARK DIST

DroneCAN and MAVLink laser distance sensor in short-, mid- and long-range variants

The USA-built ARK DIST pairs a Broadcom AFBR-S50 time-of-flight sensor with an STM32F412VG. The SR, MR and LR variants share one board and case and differ only in the sensor module.



KEY FEATURES

- Broadcom AFBR-S50 time-of-flight sensor, 850 nm Class 1 laser
- Max reported distance: SR 30 m, MR 50 m, LR 100 m
- DroneCAN and MAVLink 2 output
- 2× CAN and UART on Pixhawk-standard JST-GH, debug on JST-SH
- PX4 and ArduPilot; ROS 2 via ros2_dronecan

SPECIFICATIONS

SENSOR	
Time-of-flight module	Broadcom AFBR-S50, see Variants
Laser	850 nm, IEC 60825-1 Class 1 (eye-safe)
Transmitter beam	2° × 2°, 3.5 cm spot at 1 m
Receiver (SR, MR)	32 pixels, 12.4° × 6.2° field of view

INTERFACES	
CAN	2× 4-pin JST-GH, DroneCAN
UART	6-pin JST-GH with RTS/CTS, MAVLink 2
Debug	6-pin JST-SH, SWD and console

PROCESSOR & POWER	
MCU	STM32F412VG
Supply voltage	5 V
Supply current	78–86 mA

ENVIRONMENTAL & MECHANICAL	
Operating temperature	−20 °C to +70 °C (SR, MR)
Storage temperature	−25 °C to +85 °C (SR, MR)
Dimensions	20 × 28 × 14 mm
Weight	4 g

AT A GLANCE	
Sensor	Broadcom AFBR-S50
Variants	SR / MR / LR
Max reported distance	30 / 50 / 100 m
Update rate	20 / 15 / 10 Hz
Supply	5 V
Dimensions	20 × 28 × 14 mm
Weight	4 g
Hardware rev	Rev 1.0

COMPLIANCE & SOURCING

- Designed and manufactured in the USA
- NDAA compliant

ORDERING & RESOURCES

STORE	arkelectron.com/product/ark-dist-sr/
	arkelectron.com/product/ark-dist-mr/
	arkelectron.com/product/ark-dist-lr/
DOCS	docs.arkelectron.com/products/sensor/ark-dist
CAD	github.com/ARK-Electronics/ark-hardware

VARIANTS

	ARK DIST SR	ARK DIST MR	ARK DIST LR
Time-of-flight module	AFBR-S50LV85D	AFBR-S50LX85D	AFBR-S50EX85D
Update rate	20 Hz	15 Hz	10 Hz
Max reported distance	30 m	50 m	100 m
Min reported distance	0 m	0.1 m	0.6 m
Operating temperature	−20 °C to +70 °C	−20 °C to +70 °C	TBD
Storage temperature	−25 °C to +85 °C	−25 °C to +85 °C	TBD

RANGE

SURFACE	ARK DIST SR		ARK DIST MR		ARK DIST LR	
	BRIGHT SUN	DUSK	BRIGHT SUN	DUSK	BRIGHT SUN	DUSK
Asphalt	18 m	30 m	18 m	40 m	18 m	40 m
Grass	18 m	30 m*	18 m	50 m*	40 m	82 m

Reliable range: the farthest distance with ≥50 % of readings valid and within 5 m of the reference, at default settings. Bright sun: sun 49–61° above the horizon; a lower sun gives more range. Dusk: sun below 6°. * Limited by the maximum reported distance; the sensor was still measuring at the flight ceiling (SR 38 m, MR 54 m). Wet surfaces reduce range; open water (~2 % reflectivity) gives only short, intermittent returns.

CONNECTOR PINOUTS

CAN CONNECTOR (×2)

PIN	SIGNAL	VOLTAGE
1	5V	5.0 V
2	CAN_P	5.0 V
3	CAN_N	5.0 V
4	GND	GND

Connector: 4-pin JST-GH, Pixhawk standard. Chain additional nodes from the second CAN connector.

UART CONNECTOR

PIN	SIGNAL	VOLTAGE
1	5V_UART	5.0 V
2	USART3_RX	3.3 V
3	USART3_TX	3.3 V
4	USART3_RTS	3.3 V
5	USART3_CTS	3.3 V
6	GND	GND

Connector: 6-pin JST-GH, Pixhawk standard.

DEBUG CONNECTOR

PIN	SIGNAL	VOLTAGE
1	3.3V	3.3 V
2	USART2_TX	3.3 V
3	USART2_RX	3.3 V
4	FMU_SWDIO	3.3 V
5	FMU_SWCLK	3.3 V
6	GND	GND

Connector: 6-pin JST-SH, Pixhawk standard.

ARK-DS-ARK-DIST | Version 1.0, 2026-10-29 | Compiled from the ARK DIST Production schematic Rev 1.0 (25 Jun 2025), ARK Electronics published documentation and ARK flight measurements. Laser and optical characteristics per Broadcom. Temperature ratings per component data sheets; the time-of-flight module sets the operating range. Specifications are subject to change without notice. © 2026 ARK Electronics LLC.